

The Utility Problem

With the development and adaptation of new metering devices, such as smart meters and other IoT devices, the value which can be obtained from the deployment of these devices is rising. If data from these devices is handled properly, this data can help utilities gain a better understanding of the grid and optimize the grid based on that. In order to achieve benefits as soon as possible, it's good to start deploying these metering devices in the grid. To achieve the best outcome, it is smart to pick the locations where the deployment of these devices will have the biggest impact. Thus, the value of these devices will be maximized.

Let's take, for example, smart meters - in some service areas, the implementation of smart meters is almost complete, but in some service areas, the deployment is just beginning. In these areas, utilities have to decide what the best approach is for the deployment of these smart meters at consumption points. In some service areas, this is a legislative requirement, in others, it is a free choice - there might be some options for deployment strategies.

However, it is not only smart meters but also other IoT devices. The deployment of these devices can take place independently of the smart meter deployment strategy, but the data can be used and therefore, additional measurement points can be obtained from the grid.

Analyzing where the deployment of the IoT or smart metering device will have maximal value is a very important process since it helps get the most out of the investment.

The Data Solutions and Methods

Awesense Energy Transition Platform is very helpful in maximizing the value of IoT and smart meter deployment because it provides multiple valuable data types for such type of analysis.

Let's take the example of smart meter deployment. Each company has its own smart meter deployment strategy. Here are some examples of methods of smart meter installation:

1. By consumption.
2. According to complaints.
3. According to risk analysis - e.g. losses.

Description of the installation methods:

1. By consumption:
 - a. This method takes advantage of the fact that the top 20% of customers consume about 80% of all energy (this is very well known as the Pareto rule). Therefore it is appropriate to focus on these customers at the beginning. This method is very efficient because, after a very short time, you have an overview of most of the flows in the grid. A very small part is left to estimate the remaining consumption, making the estimation more accurate.

- b. Since the utility does not have time series data yet, it works with meter readings and the readings in kWh are exported from the database and then the average annual value of each consumption point is calculated. The consumption points are ranked from the highest annual consumption to the lowest and the highest consumption points can be marked on a map, also a colour scale can be used according to the consumption of all consumption points to ensure for example, that no parts of the grid have been ignored.
- 2. According to complaints:
 - a. If there are more customer complaints (for example, about power quality) at some points in the grid than at others, it is advisable to install smart meters at these points that are able to partially measure this (power quality - voltage value).
- 3. According to risk analysis - e.g. losses:
 - a. This method is suitable if the utility has some problem in the grid. The grid is divided into segments based on a selected parameter or combination of parameters and the resulting value is calculated for each segment. This method can be used in many ways - for example, to find suitable areas of the grid for:
 - i. innovation,
 - ii. when knowledge of the area is poor,
 - iii. with high crime rates,
 - iv. when there is a mismatch in consumption,
 - v. zero or significantly declining consumption
 - vi. etc.

With True Grid Intelligence (TGI) web application, this analysis can be as well performed directly in TGI using the Segment Analysis feature. The applied analysis in TGI can also take into account all three of the above methods. By combining multiple factors, the resulting analysis may be the most efficient. You can very easily add another installation method and use it alone or with the other methods mentioned above. Defining the parameters is simple and intuitive.

Thanks to the synchronized data in the Awesense Energy Data Model (EDM), we are able to visualize the data on a map for better evaluation. If the result is only displayed in tables, the added value of this analysis may not be complete. When displayed on a map, it is very easy to detect places where new meters are completely missed. However, it would be useful, for example, to have all areas at least partially covered - e.g. to determine the voltage parameters of an area, etc.

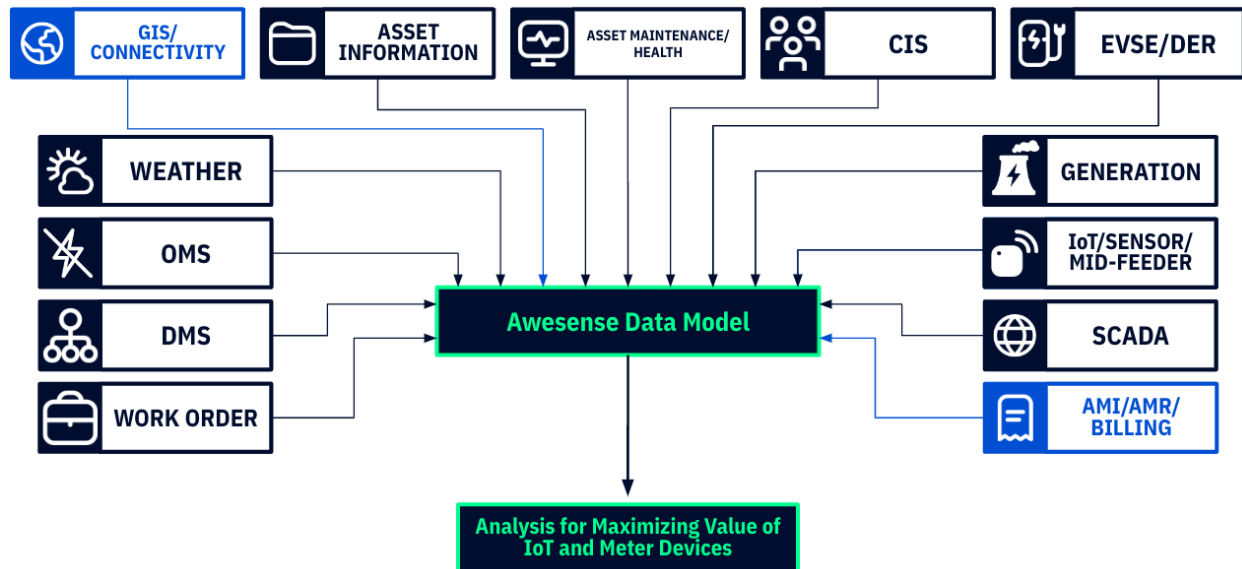
Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.

Analysis for Maximizing Value of IoT and Meter Devices



[UC25-01] - Awesense Energy Transition Platform Use Case



Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. GitHub link to the Jupyter notebook can be provided upon request.
2. TGI Segment Analyzer - Please speak to Awesense about access to the sandbox environment.